

Near-Exact Filtering and Parameter Inference for Robust Location Models

Particle Filtering and Near-Exact Benchmarking for Robust
State-Space Models

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Motivation

Problem: filter a latent signal μ_t from heavy-tailed observations y_t .

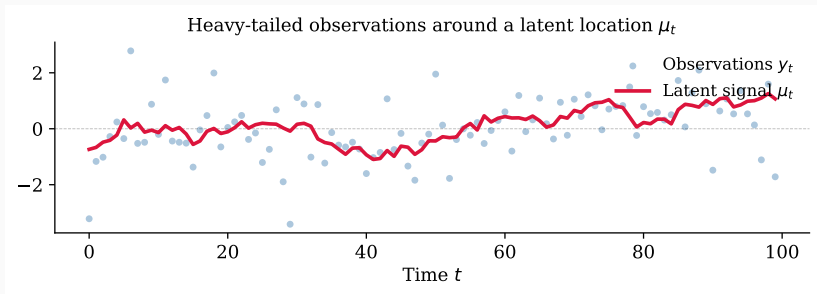
Two approaches — main method and tractable baseline:

- **Particle filtering:** reconstructs the full posterior $p(\mu_t \mid y_{0:t})$ of model (B) — the main inferential method
- **Score-driven (GAS):** fast, tractable approximation — used as an observation-driven baseline

Built on the ℓ_ν (limit-Gaussian) innovation family (Luati, 2025):
sub-Gaussian for finite ν , Gaussian as $\nu \rightarrow \infty$.

Three aims: (A) near-exact benchmarking (B) Kalman-like regime
(C) identifiability of ν

Motivation



The Two Models

(A) Score-driven — *tractable observation-driven baseline* [Harvey & Luati, 2014]

$$p(y_t \mid \mathcal{F}_{t-1}, \mu_t) \sim t_\nu(\mu_t, \sigma), \quad \mu_t = \phi \mu_{t-1} + \kappa x_{t-1}$$

$$x_t = \frac{y_t - \mu_t}{1 + \frac{(y_t - \mu_t)^2}{\nu \sigma^2}} \quad \leftarrow \text{robust scaled score}$$

(B) State-space robust location [Density2, Luati 2025]

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim t_\nu(0, \sigma) \\ \mu_t = \phi \mu_{t-1} + \eta_t, \quad \eta_t \sim \ell_\nu(0, \varpi)$$

ℓ_ν = limit-Gaussian: bounded support for finite ν , $\ell_\nu \rightarrow \mathcal{N}$ as $\nu \rightarrow \infty$.

Target: $p(\mu_t \mid y_{0:t})$.

Aim A: Near-Exact Benchmarking

Q: How accurately do particle methods reconstruct $p(\mu_t \mid y_{0:t})$ for model (B), across $(\nu, \sigma, \varpi, \phi)$ and under outliers?

Approach: deterministic 1D grid/quadrature filter as near-exact reference for $p(\mu_t \mid y_{0:t})$.

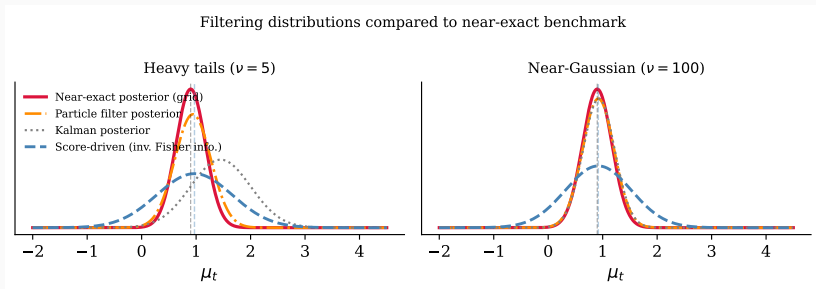
Methods evaluated:

1. Bootstrap PF & Auxiliary PF (*primary reconstruction methods*)
2. PMMH-based parameter learning
3. Score-driven filter (model A) — *observation-driven baseline*

Metrics: filtering accuracy, predictive scoring rules, uncertainty quantification, computational efficiency.

Sweeps: $\nu \in \{5, 10, 30, 100, 1000\}$, multiple (σ, ϖ, ϕ) regimes, with/without additive outliers.

Aim A: Filter Distributions Compared



(Schematic illustration — actual filter distributions will be computed against the 1D grid benchmark.)

Score-driven is displayed as $\mathcal{N}(\hat{\mu}_t, \mathcal{F}^{-1})$, where $\mathcal{F} = (\nu+1)/[\sigma^2(\nu+3)]$ is the Fisher information of t_ν for location. PF approximates the true posterior of model (B); Kalman computes the exact posterior under a Gaussian approximation. Left ($\nu = 5$): PF targets the posterior directly; score-driven, as a reduced-form approximation, is evaluated against the near-exact reference; Kalman is pulled toward the outlier. Right ($\nu = 100$): all converge.

Aim B: The Kalman-Like Regime

Q: For what ν do simpler approximations — Kalman and score-driven — become adequate substitutes for the exact non-Gaussian filter?

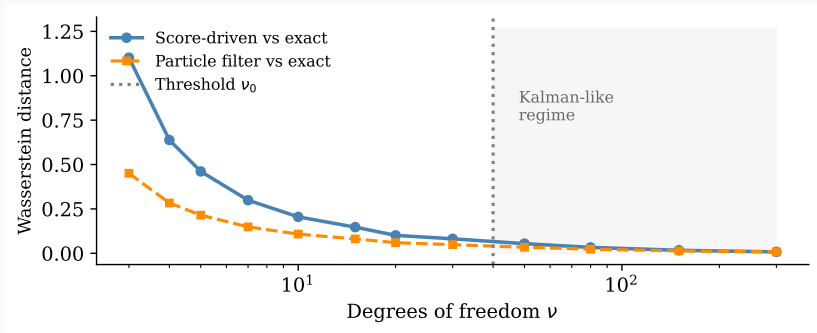
Approach:

- Quantify **Wasserstein distance** and **moment gaps** between non-Gaussian and Kalman filtering distributions, using the 1D benchmark
- Define operational threshold ν_0 : above it, Gaussian approximation is safe

Why it matters:

- Gives practitioners a concrete criterion for when simpler approximations (Kalman, score-driven) are acceptable
- The same ν_0 directly informs the identifiability problem in Aim C

Aim B: Distance to Exact Benchmark vs ν



(Schematic illustration — Wasserstein distances to be computed from the 1D grid benchmark.)

Aim C: Identifiability of ν

Q: As $\nu \rightarrow \infty$, the likelihood flattens - how bad is it and what can we do?

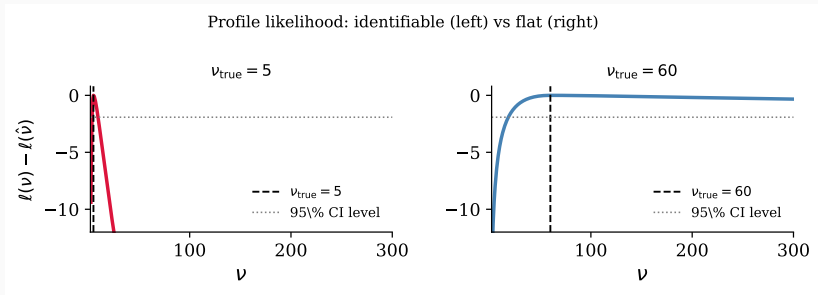
Approach:

- **Profile likelihood** $\ell(\nu)$ across true ν values
- **Curvature diagnostics** under reparametrisation, e.g. $\log(\nu-2)$
- **Bayesian posteriors** for ν via PMMH, varying ν_{true}

Deliverable: characterisation of the flat-likelihood phenomenon + practical remedies:

- Reparametrisation
- Informative priors
- Threshold-to-Gaussian

Aim C: Profile Likelihood - Identifiable vs Flat



(Schematic illustration — profile likelihood curves to be computed from simulated data.)

Optional Extension: Exact Filtering via Dual Process

Ambitious goal (Jul–Aug, only if core aims complete):

Does model (B) with ℓ_ν innovations admit a *computable exact filter*?

Background [Kon Kam King, Papaspiliopoulos & Ruggiero, 2021]:

If the signal μ_t has a *dual process* - an auxiliary process embedded in its time reversal - then:

- all filtering distributions are **finite mixtures** of conjugate densities
- mixtures are updated via **exact recursive formulae**
- can outperform particle filters in classes of models where the dual structure exists

The question: does the $\ell_\nu(0, \varpi)$ state transition in model (B) admit such a dual structure, yielding a computable finite-dimensional filter?

Why it matters: if yes, this would provide a third benchmark alongside the grid filter and particle filter - going beyond approximation to exact inference for this class of robust location models.

Implementation

1. 1D grid filter

Quadrature prediction/update
using $\ell_\nu(0, \varpi)$ innovation law

2. Particle methods

Bootstrap PF and Auxiliary PF;
ESS and log-likelihood variance
diagnostics

3. Parameter inference

MLE / profile $\ell(\nu)$ - model (A)
PMMH - model (B)

Data

*A latent location process with
heavy-tailed observations:*

$$y_t \sim t_\nu(\mu_t, \sigma), \quad \eta_t \sim \ell_\nu(0, \varpi)$$

Tail thickness ν and signal-to-noise
controlled, spanning heavy-tailed to
near-Gaussian.

Framework applies to any time series
with analogous robust-location
structure.

Timeline and Expected Contributions

Milestones

- Mar–Apr** 1D grid benchmark,
PF/APF diagnostics,
experiment harness
- May–Jun** PMMH,
profile-curvature
analysis, Kalman
threshold ν_0
- Jul–Aug** Results, empirical
validation, writing;
dual-process exact filter
(optional)

Expected contributions

- Near-exact 1D benchmark under ℓ_ν innovations
- Evaluation of particle filter performance and score-driven approximation error against the near-exact benchmark
- Operational characterisation of the $\nu \rightarrow \infty$ Kalman-like regime
- Practical guidance on ν identifiability

Thank you - happy to take questions.